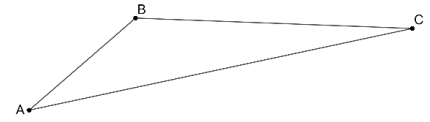


Geometry
Unit 11
Lesson 5 Homework

Name **Answer Key**
Date _____
Period _____

For questions 1 through 4, $\triangle ABC$ is shown, where $m\angle A = (3x)^\circ$, $m\angle B = (7x + 10)^\circ$, and $m\angle C = (2x + 10)^\circ$.



1. Find the value of x .

$$\begin{aligned} 3x + (7x + 10) + (2x + 10) &= 180 \\ 12x + 20 &= 180 \\ 12x &= 160 \\ x &= 13\frac{1}{3} \end{aligned}$$

2. What is the $m\angle A$?

$$m\angle A = 3\left(13\frac{1}{3}\right) = 40^\circ$$

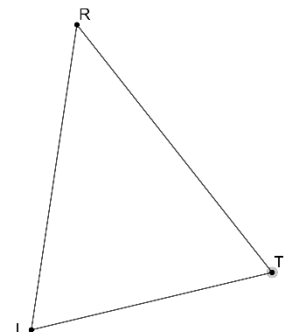
3. What is the $m\angle B$?

$$m\angle B = 7\left(13\frac{1}{3}\right) + 10 = 103\frac{1}{3}^\circ$$

4. What is the $m\angle C$?

$$m\angle C = 2\left(13\frac{1}{3}\right) + 10 = 36\frac{2}{3}^\circ$$

For questions 5 and 6, $\triangle LRT$ is shown, where $m\angle L = m\angle T = (5x)^\circ$ and $m\angle R = (2x)^\circ$.



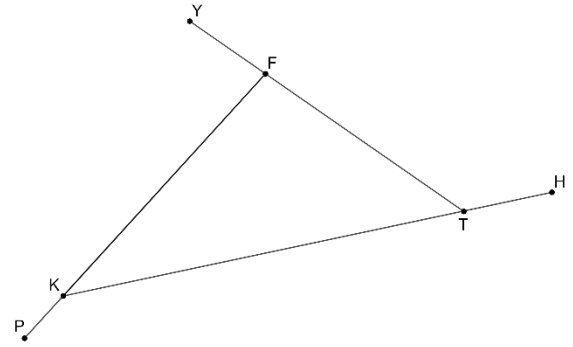
5. Find the value of x .

$$\begin{aligned} 5x + 5x + 2x &= 180 \\ 12x &= 180 \\ x &= 15 \end{aligned}$$

6. What is the $m\angle L$?

$$m\angle L = 5(15) = 75^\circ$$

For questions 7 through 10, $\triangle FKT$ is shown, where $\angle KFY$, $\angle PKT$, and $\angle FTH$ are exterior angles.
 $m\angle FTH = (8x + 10)^\circ$, $m\angle TFK = 88^\circ$, and
 $m\angle PKT = (16x - 30)^\circ$.



7. Find the value of x .

$$\begin{aligned}
 m\angle KFY &= 180 - 88 = 92^\circ \\
 92 + (16x - 30) + (8x + 10) &= 360 \\
 24x + 72 &= 360 \\
 24x &= 288 \\
 x &= 12
 \end{aligned}$$

8. Find $m\angle KTF$.

$$m\angle KTF = 180 - [8(12) + 10] = 74^\circ$$

9. Find $m\angle PKT$.

$$m\angle PKT = 16(12) - 30 = 162^\circ$$

10. Find $m\angle YFK$.

$$m\angle YFK = 180 - 88 = 92^\circ$$